

Operations Research

Lecture 4: Duality of Linear Programs

A textbook treatment

Based on the lecture notes of El Mehdi Saad

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How to read this book

Each concept in this book is presented in the same rhythm.

- 1. Motivation and intuition.** Why do we care? What problem does the idea solve, and what picture should you hold in your head?
- 2. Formal statement.** The precise definition or theorem, set in a coloured box so it is easy to find when revising.
- 3. Proof.** Every claim is proved. Longer proofs are split into labelled steps, and the one idea that makes the proof work is isolated in a *Key insight* box.
- 4. Interpretation and discussion.** What the result really says, where it applies, and which misreadings to avoid.
- 5. Examples.** From “trivially small” to “genuinely useful.”
- 6. Techniques, classic problems and gotchas.** Computational shortcuts, proof patterns, exam favourites with one-line solutions, and the mistakes everyone makes once.

Margin notes carry the informal, “say it out loud” version of what is happening in the main text. Every major section closes with a summary box.

Prerequisites. Comfort with vectors, matrices, systems of linear equations, row reduction and the notions of rank and linear independence. Chapter 2 recalls exactly what is needed as it goes.

Chapter 4

Duality of Linear Programs

4.1 From a Feasible Plan to a Certificate: Bounding the Optimum

Motivation

Chapter 3 showed how to *find* an optimal plan. This chapter asks a different question: how do you *know* a plan is optimal, or how far from optimal it is, without finding the optimum first? A workshop that already runs a feasible production plan does not necessarily want to re-solve its whole problem; it wants to know how much profit it might be leaving on the table. Answering that requires an *upper bound* on the best possible profit. The surprising fact is that the search for the best such bound is itself a linear program, the *dual*, and that its optimal value equals the original optimum exactly. Everything in this chapter grows out of one running example.

A feasible plan tells you what you can get. A certificate tells you what you cannot beat. Optimisation is finished when the two numbers agree.

Example 4.1.1 (The workshop). A workshop chooses two divisible activity levels $x_1, x_2 \geq 0$. Three limited resources constrain production:

$$\begin{aligned} & \max \quad 2x_1 + 3x_2 \\ \text{s.t.} \quad & 4x_1 + 8x_2 \leq 12, \quad 2x_1 + x_2 \leq 3, \quad 3x_1 + 2x_2 \leq 4, \quad x_1, x_2 \geq 0. \end{aligned}$$

In the matrix notation of Definition 2.1.1,

$$A = \begin{bmatrix} 4 & 8 \\ 2 & 1 \\ 3 & 2 \end{bmatrix}, \quad b = \begin{bmatrix} 12 \\ 3 \\ 4 \end{bmatrix}, \quad c = \begin{bmatrix} 2 \\ 3 \end{bmatrix}.$$

Consider the plan $\hat{x} = (0, 3/2)$. Its resource use is $4(0) + 8(3/2) = 12 \leq 12$, $2(0) + 3/2 = 3/2 \leq 3$ and $3(0) + 2(3/2) = 3 \leq 4$; both coordinates are nonnegative, so the plan is feasible, with profit $2(0) + 3(3/2) = 9/2$.

Feasibility says nothing about quality. Write p^* for the (unknown) maximum profit. Because $\hat{x}$ is feasible, $9/2 \leq p^*$, and the *improvement still possible* is $p^* - 9/2 \geq 0$: zero if $\hat{x}$ is optimal, positive otherwise. We would like to bound this quantity before finding an optimal solution.

Upper bounds from the constraints

Suppose we can prove that every feasible plan has profit at most U , i.e. $2x_1 + 3x_2 \leq U$ for every feasible x . Then

$$\frac{9}{2} \leq p^* \leq U \quad \implies \quad 0 \leq p^* - \frac{9}{2} \leq U - \frac{9}{2}.$$

If $U = 9/2$ the plan is optimal; if U is close to $9/2$ the plan is close to optimal. The bound limits the possible loss; it need not give the exact loss.

Example 4.1.2 (A first bound, by eye). Divide the first resource inequality by 2: $4x_1 + 8x_2 \leq 12$ gives $2x_1 + 4x_2 \leq 6$. Since $x_2 \geq 0$, every feasible point satisfies

$$\underbrace{2x_1 + 3x_2}_{\text{profit}} \leq 2x_1 + 4x_2 \leq 6.$$

Hence $9/2 \leq p^* \leq 6$ and $0 \leq p^* - 9/2 \leq 3/2$: the plan is at most 1.5 profit units below the optimum.

Example 4.1.3 (A better bound: combine two constraints). Take one third of each of the first two inequalities and add:

$$\frac{1}{3}(4x_1 + 8x_2) + \frac{1}{3}(2x_1 + x_2) \leq \frac{1}{3} \cdot 12 + \frac{1}{3} \cdot 3 \quad \implies \quad 2x_1 + 3x_2 \leq 5.$$

The left-hand side is *exactly* the profit, so $p^* \leq 5$ and $0 \leq p^* - 9/2 \leq 1/2$. The plan is now guaranteed to be within 0.5 of optimal.

The two bounds came from different choices of multipliers:

Constraint 1	Constraint 2	Constraint 3	Bound
1/2	0	0	6
1/3	1/3	0	5
y_1	y_2	y_3	?

The natural next step is to let the multipliers themselves become decision variables and ask: *which nonnegative weights give the smallest upper bound obtainable this way?*

Multiply, add, dominate

Proposition 4.1.4 (Nonnegative combinations of constraints are valid bounds). *In Example 4.1.1, let $y_1, y_2, y_3 \geq 0$ satisfy*

$$4y_1 + 2y_2 + 3y_3 \geq 2 \quad \text{and} \quad 8y_1 + y_2 + 2y_3 \geq 3.$$

Then every feasible x satisfies $2x_1 + 3x_2 \leq 12y_1 + 3y_2 + 4y_3$.

Proof. **Step 1 (multiply).** Nonnegative weights preserve $\leq$ (Proposition 2.1.2):

$$y_1(4x_1 + 8x_2) \leq 12y_1, \quad y_2(2x_1 + x_2) \leq 3y_2, \quad y_3(3x_1 + 2x_2) \leq 4y_3.$$

Step 2 (add and regroup by variable). Summing,

$$(4y_1 + 2y_2 + 3y_3)x_1 + (8y_1 + y_2 + 2y_3)x_2 \leq 12y_1 + 3y_2 + 4y_3.$$

Step 3 (dominate the profit). Because $x_1, x_2 \geq 0$, a coefficient that is at least the profit coefficient can only make the left-hand side larger: $2x_1 \leq (4y_1 + 2y_2 + 3y_3)x_1$ and $3x_2 \leq (8y_1 + y_2 + 2y_3)x_2$. Adding these two inequalities to the one in Step 2 gives the claim. $\square$

Key insight

Two sign conditions make the argument work, and they will reappear in every proof of this chapter: multipliers $y_i \geq 0$ so that the resource inequalities keep their direction, and variables $x_j \geq 0$ so that a larger coefficient means a larger term. Each condition is used exactly once.

The search for the best bound is an LP

Among all weights that make the argument valid, we want the smallest bound:

$$\begin{aligned} \min \quad & 12y_1 + 3y_2 + 4y_3 \\ \text{s.t.} \quad & 4y_1 + 2y_2 + 3y_3 \geq 2, \quad 8y_1 + y_2 + 2y_3 \geq 3, \quad y_1, y_2, y_3 \geq 0. \end{aligned}$$

This is a new linear program, the *dual problem*. Its feasible solutions are certificates of upper bounds for the original LP. Side by side:

Primal: choose activity levels	Dual: choose constraint weights
$\max \quad 2x_1 + 3x_2$ $\text{s.t.} \quad 4x_1 + 8x_2 \leq 12$ $2x_1 + x_2 \leq 3$ $3x_1 + 2x_2 \leq 4$ $x_1, x_2 \geq 0$	$\min \quad 12y_1 + 3y_2 + 4y_3$ $\text{s.t.} \quad 4y_1 + 2y_2 + 3y_3 \geq 2$ $8y_1 + y_2 + 2y_3 \geq 3$ $y_1, y_2, y_3 \geq 0$
A feasible plan gives a <i>lower</i> bound on the maximum profit.	Feasible weights give an <i>upper</i> bound on the maximum profit.

For every feasible pair x, y : $2x_1 + 3x_2 \leq p^* \leq 12y_1 + 3y_2 + 4y_3$.

Definition 4.1.5 (The canonical primal–dual pair). Let $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, $c \in \mathbb{R}^n$. The *primal* and *dual* linear programs are

$$\begin{aligned} (P) \quad & \max \quad c^\top x \quad \text{s.t.} \quad Ax \leq b, \quad x \geq 0, \\ (D) \quad & \min \quad b^\top y \quad \text{s.t.} \quad A^\top y \geq c, \quad y \geq 0. \end{aligned}$$

(*P*) has n variables and m inequalities; (*D*) has m variables and n inequalities. One primal constraint gives one dual variable; one primal variable gives one dual constraint. We write p^* and d^* for their optimal values when these exist.

Interpretation & discussion

Why does the matrix get transposed? In the workshop,

$$A = \begin{bmatrix} 4 & 8 \\ 2 & 1 \\ 3 & 2 \end{bmatrix}, \quad y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}, \quad A^\top y = \begin{bmatrix} 4y_1 + 2y_2 + 3y_3 \\ 8y_1 + y_2 + 2y_3 \end{bmatrix}.$$

Primal column j lists the resource use of activity j ; dual row j is the *weighted*

Rows become variables, columns become constraints. The transpose is not a convention: it is what “regroup by variable” does to the coefficients.

resource use of activity j , which must dominate its profit c_j . In indices, the j -th dual constraint reads $\sum_{i=1}^m a_{ij}y_i \geq c_j$, $j = 1, \dots, n$: the same numbers a_{ij} , read down a column instead of along a row. The objective $b^\top y$ is the weighted sum of capacities, i.e. the bound itself.

Technique: Writing the canonical dual in thirty seconds

- Count: m primal rows $\Rightarrow m$ dual variables $y_1, \dots, y_m \geq 0$; n primal variables $\Rightarrow n$ dual constraints.
- Dual objective: the primal right-hand sides b become the cost vector; max becomes min.
- Dual constraint j : read primal *column* j top to bottom, dot it with y , and require $\geq c_j$.
- Dimension check: $A^\top$ is $n \times m$ and y is $m \times 1$, so $A^\top y$ is $n \times 1$, matching c .

Summary: Bounding the optimum

- A feasible plan gives a lower bound on p^* ; a proof that no plan beats U gives an upper bound, and $U - (\text{plan value})$ bounds the possible loss.
- Multiplying constraints by nonnegative weights, adding, and choosing weights whose regrouped coefficients dominate the objective produces a valid upper bound (Proposition 4.1.4).
- Minimising that bound over all valid weights is a linear program: the dual (Definition 4.1.5). Rows become variables, columns become constraints, and A becomes $A^\top$.

4.2 Weak Duality: Every Certificate Is Valid

Motivation

Proposition 4.1.4 was proved for three specific constraints. The general statement, for any primal–dual pair, is the first theorem of duality theory. It is short, but it is the engine behind optimality certificates, the duality gap, and the primal–dual case analysis of Section 4.6. Lemma 1.3.10 already stated it in Chapter 1; here we name it, reprove it, and draw every consequence.

Weak duality is the “always safe” half of the story: whatever dual-feasible weights you pick, the bound they give is never wrong.

Theorem 4.2.1 (Weak duality). *For every primal-feasible x and every dual-feasible y of the canonical pair of Definition 4.1.5,*

$$c^\top x \leq b^\top y.$$

Proof. Two steps, each using one sign condition.

$$\begin{aligned} c^\top x &\leq (A^\top y)^\top x = y^\top Ax && \text{because } A^\top y \geq c \text{ and } x \geq 0, \\ y^\top Ax &= y^\top (Ax) \leq y^\top b = b^\top y && \text{because } Ax \leq b \text{ and } y \geq 0. \end{aligned}$$

Both inequalities are instances of Proposition 2.1.2: $u \leq v$ and $w \geq 0$ imply $w^\top u \leq w^\top v$. Chain them. $\square$

Key insight

The quantity $y^\top Ax$ sits between the two objectives. It can be read in two ways: as $(A^\top y)^\top x$, a dual row applied to x , or as $y^\top (Ax)$, the weights applied to the resource use. Every duality proof in this chapter inserts this middle term.

Corollary 4.2.2 (Bounds and the duality gap). *Let x be primal-feasible and y dual-feasible.*

- (a) $c^\top x \leq p^*$ and $d^* \leq b^\top y$ whenever these optimal values exist; moreover $p^* \leq b^\top y$ and $c^\top x \leq d^*$.
- (b) The duality gap $b^\top y - c^\top x$ is nonnegative, and $p^* - c^\top x \leq b^\top y - c^\top x$: the gap bounds the loss of the plan x .

Proof. (a) $c^\top x \leq p^*$ and $d^* \leq b^\top y$ are the definitions of maximum and minimum. For $p^* \leq b^\top y$: every feasible x' satisfies $c^\top x' \leq b^\top y$ by Theorem 4.2.1, so the maximum does too. Symmetrically for $c^\top x \leq d^*$. (b) Nonnegativity is the theorem; subtract $c^\top x$ from $p^* \leq b^\top y$. $\square$

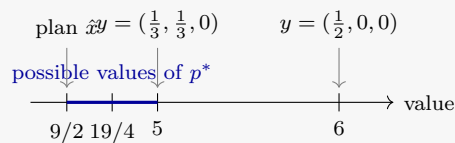
Theorem 4.2.3 (A matching pair proves optimality). *If x is primal-feasible, y is dual-feasible and $c^\top x = b^\top y$, then x is optimal for (P), y is optimal for (D), and $p^* = d^* = c^\top x$.*

Proof. For every primal-feasible x' , $c^\top x' \leq b^\top y = c^\top x$ by Theorem 4.2.1; so x is optimal and $p^* = c^\top x$. For every dual-feasible y' , $b^\top y' \geq c^\top x = b^\top y$; so y is optimal and $d^* = b^\top y$. $\square$

Interpretation & discussion

Theorem 4.2.3 is a *verification* procedure that costs two matrix–vector products and one comparison. It does not require any algorithm, any tableau, or any knowledge of how x and y were found. On an exam, “feasible x , feasible y , equal values” is a complete proof of optimality. The theorem says nothing about whether such a pair *exists*; that is the job of strong duality (Section 4.3).

Example 4.2.4 (Trapping the optimum of the workshop). The plan $\hat{x} = (0, 3/2)$ has value $9/2$. The weights $y = (1/3, 1/3, 0)$ are dual-feasible ($4/3 + 2/3 = 2 \geq 2$, $8/3 + 1/3 = 3 \geq 3$, all ≥ 0) with value $12/3 + 3/3 = 5$. Hence $9/2 \leq p^* \leq 5$.



A better certificate, $y = (5/16, 0, 1/4)$, is also dual-feasible: $4 \cdot \frac{5}{16} + 3 \cdot \frac{1}{4} = \frac{5}{4} + \frac{3}{4} = 2$ and $8 \cdot \frac{5}{16} + 2 \cdot \frac{1}{4} = \frac{5}{2} + \frac{1}{2} = 3$, with value $12 \cdot \frac{5}{16} + 4 \cdot \frac{1}{4} = \frac{15}{4} + 1 = \frac{19}{4}$. So $9/2 \leq p^* \leq 19/4$, and the duality gap of the pair $(\hat{x}, y)$ is $19/4 - 9/2 = 1/4$.

Example 4.2.5 (A matching pair for the workshop). Take $x^* = (1/2, 5/4)$ and $y^* = (5/16, 0, 1/4)$. Primal feasibility:

$$Ax^* = \begin{bmatrix} 4 \cdot \frac{1}{2} + 8 \cdot \frac{5}{4} \\ 2 \cdot \frac{1}{2} + \frac{5}{4} \\ 3 \cdot \frac{1}{2} + 2 \cdot \frac{5}{4} \end{bmatrix} = \begin{bmatrix} 12 \\ 9/4 \\ 4 \end{bmatrix} \leq \begin{bmatrix} 12 \\ 3 \\ 4 \end{bmatrix}, \quad x^* \geq 0.$$

Dual feasibility was checked in Example 4.2.4: $A^\top y^* = (2, 3)^\top = c$ and $y^* \geq 0$. Values:

$$c^\top x^* = 2 \cdot \frac{1}{2} + 3 \cdot \frac{5}{4} = 1 + \frac{15}{4} = \frac{19}{4}, \quad b^\top y^* = \frac{15}{4} + 0 + 1 = \frac{19}{4}.$$

By Theorem 4.2.3 both are optimal and $p^* = d^* = 19/4$. The exact loss of the initial plan $\hat{x}$ is now known: $19/4 - 9/2 = 1/4$.

Technique: Reading a dual solution as a proof

A dual-feasible y is a recipe: “multiply constraint i by y_i and add.” Writing the recipe out makes the bound visible to anyone. For $y^* = (5/16, 0, 1/4)$:

$$\frac{5}{16}(4x_1 + 8x_2) + \frac{1}{4}(3x_1 + 2x_2) = 2x_1 + 3x_2 \leq \frac{5}{16} \cdot 12 + \frac{1}{4} \cdot 4 = \frac{19}{4}.$$

When $A^\top y = c$ exactly (as here), the recipe reproduces the objective and the bound is read off directly; when $A^\top y > c$ in some coordinate, one more line “and $x \geq 0$ ” finishes the argument.

Classic problems

- For $\max 3x_1 + 2x_2$ s.t. $x_1 + x_2 \leq 4$, $2x_1 + x_2 \leq 6$, $x \geq 0$, is $x = (2, 2)$ optimal? Add the two constraints: $3x_1 + 2x_2 \leq 10$; $x = (2, 2)$ is feasible with value 10. Yes, by Theorem 4.2.3 with $y = (1, 1)$.
- x has value 7 and some dual-feasible y has value 7.3. What do you know? $7 \leq p^* \leq 7.3$; the loss of x is at most 0.3; nothing more without further work.
- Can a dual-feasible y have $b^\top y < c^\top x$ for a primal-feasible x ? No: that would contradict Theorem 4.2.1. If it “happens” in a calculation, one of the two vectors is infeasible.
- Is $y = (1/2, 0, 0)$ a worse certificate than $y = (1/3, 1/3, 0)$ in the workshop? It gives bound $6 > 5$: valid but weaker. Both are dual-feasible; the dual objective ranks them.

Gotcha

- Weak duality compares a feasible x with a feasible y . A “bound” computed from weights that violate $A^\top y \geq c$ or have a negative component is not a bound at all.
- The gap $b^\top y - c^\top x$ bounds the loss of x but does not equal it: in Example 4.2.4, with $y = (1/3, 1/3, 0)$ the gap was $1/2$ while the true loss of $\hat{x}$ is $1/4$.
- Equal values prove optimality only together with *both* feasibilities.

Checking $c^\top x = b^\top y$ for an infeasible y proves nothing.

Summary: Weak duality

- $c^\top x \leq y^\top Ax \leq b^\top y$ for every primal-feasible x and dual-feasible y (Theorem 4.2.1); for maximisation, x gives a lower bound and y an upper bound.
- The duality gap $b^\top y - c^\top x \geq 0$ bounds how far x is from optimal (Corollary 4.2.2).
- Feasible x , feasible y , equal values $\Rightarrow$ both optimal (Theorem 4.2.3): a two-line proof anyone can check.

4.3 Strong Duality: The Best Bounds Meet

Motivation

In Example 4.2.5 the best plan and the best certificate had the same value, 19/4. Was that luck? Strong duality says no: whenever a canonical LP has a finite optimum, the dual has one too, and the two values coincide. This is the deepest theorem of the chapter and the one the slides state without proof. We prove it from scratch. The route is through a classical alternative theorem, the Farkas lemma, which itself follows from an elimination procedure that is nothing but “combine inequalities with nonnegative weights” used systematically.

Weak duality: every certificate is valid. Strong duality: some certificate is exact. The gap between the best plan and the best bound is zero.

Theorem 4.3.1 (Strong duality). *If the primal (P) of Definition 4.1.5 is feasible and its objective is bounded above on the feasible set, then both (P) and (D) have optimal solutions x^* , y^* , and*

$$\max\{c^\top x : Ax \leq b, x \geq 0\} = \min\{b^\top y : A^\top y \geq c, y \geq 0\}.$$

The feasible region need not be bounded; only the objective value must be. The proof occupies the rest of this section. We first show that an optimal x^* exists, then build y^* .

Step 0: the primal optimum is attained

Lemma 4.3.2 (Existence of a primal optimum). *Under the hypotheses of Theorem 4.3.1, (P) has an optimal solution.*

Proof. Add slacks (Definition 2.2.3) to obtain the equational form $\max c^\top x$ s.t. $Ax + s = b$, $(x, s) \geq 0$, whose constraint matrix $[A \ I_m]$ has rank m . By Theorem 2.2.4 this problem is feasible, and its objective is bounded above by the same bound, since feasible (x, s) correspond to feasible x with equal objective. Theorem 2.5.2 then gives an optimal solution (x^*, s^*) , and x^* is optimal for (P) by the same correspondence. $\square$

Step 1: eliminating a variable from a system of inequalities

Lemma 4.3.3 (Fourier–Motzkin elimination). *Let $Ax \leq b$ be a system of m linear inequalities in the variables $x_1, \dots, x_n$ with $n \geq 1$. There is a system $A'x' \leq b'$ in the variables $x' = (x_2, \dots, x_n)$ such that*

- (a) *every row (a'_k, b'_k) of $[A' \ b']$ is a nonnegative combination of rows of $[A \ b]$ whose x_1 -coefficient is zero, i.e. $a'_k = \sum_i \lambda_{ki} a_i$, $b'_k = \sum_i \lambda_{ki} b_i$ with all $\lambda_{ki} \geq 0$ and $\sum_i \lambda_{ki} a_{i1} = 0$;*
- (b) *$Ax \leq b$ has a solution if and only if $A'x' \leq b'$ has a solution.*

Proof. Step 1 (sort the rows by the sign of the x_1 -coefficient). Let $P = \{i : a_{i1} > 0\}$, $N = \{i : a_{i1} < 0\}$, $Z = \{i : a_{i1} = 0\}$. Write row i as $a_{i1}x_1 + \bar{a}_i^\top x' \leq b_i$ where $\bar{a}_i$ collects the remaining coefficients. For $i \in P$ divide by $a_{i1} > 0$: $x_1 \leq u_i(x') := (b_i - \bar{a}_i^\top x')/a_{i1}$. For $k \in N$ divide by $a_{k1} < 0$, which reverses the inequality (Proposition 1.3.4): $x_1 \geq \ell_k(x') := (b_k - \bar{a}_k^\top x')/a_{k1}$.

Step 2 (define the new system). Take all rows of Z unchanged, and for every pair $(k, i) \in N \times P$ the inequality $\ell_k(x') \leq u_i(x')$, which after clearing denominators reads

$$\left(\frac{1}{a_{i1}} \bar{a}_i - \frac{1}{a_{k1}} \bar{a}_k \right)^\top x' \leq \frac{1}{a_{i1}} b_i - \frac{1}{a_{k1}} b_k.$$

This row is $\frac{1}{a_{i1}}(\text{row } i) + \frac{1}{-a_{k1}}(\text{row } k)$: both multipliers are positive, and the x_1 -coefficient is $\frac{a_{i1}}{a_{i1}} + \frac{a_{k1}}{-a_{k1}} = 1 - 1 = 0$. Rows of Z are combinations with a single multiplier 1. This proves (a). (If P or N is empty, the new system consists of the Z rows only; if Z is also empty, it is the empty system, which every x' satisfies.)

Step 3 (solutions go down). If $x = (x_1, x')$ satisfies $Ax \leq b$, then x' satisfies every Z row, and for each pair, $\ell_k(x') \leq x_1 \leq u_i(x')$, so $\ell_k(x') \leq u_i(x')$. Hence x' solves the new system.

Step 4 (solutions come back up). If x' solves the new system, then $\max_{k \in N} \ell_k(x') \leq \min_{i \in P} u_i(x')$ (every pair was checked). Choose x_1 in this interval; if $N = \emptyset$ take $x_1 = \min_{i \in P} u_i(x')$, if $P = \emptyset$ take $x_1 = \max_{k \in N} \ell_k(x')$, if both are empty take $x_1 = 0$. Then $x_1 \leq u_i(x')$ for all $i \in P$ and $x_1 \geq \ell_k(x')$ for all $k \in N$, which are the P and N rows; the Z rows hold by assumption. So (x_1, x') solves $Ax \leq b$. Together with Step 3 this proves (b). $\square$

Key insight

Eliminating x_1 means: every lower bound on x_1 must be at most every upper bound on x_1 . Writing those comparisons out is exactly “add a positive multiple of a row with $a_{i1} > 0$ to a positive multiple of a row with $a_{k1} < 0$ so that x_1 cancels.” The procedure is a machine that manufactures nonnegative combinations.

Step 2: the Farkas lemma

Theorem 4.3.4 (Farkas lemma, inequality form). *Let $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$. Exactly one of the following holds:*

- (i) *there is $x \in \mathbb{R}^n$ with $Ax \leq b$;*
- (ii) *there is $y \in \mathbb{R}^m$ with $y \geq 0$, $A^\top y = 0$ and $b^\top y < 0$.*

Proof. Not both. If x and y satisfied (i) and (ii), then $0 = (A^\top y)^\top x = y^\top (Ax) \leq y^\top b < 0$ (the inequality by $Ax \leq b$, $y \geq 0$ and Proposition 2.1.2), a contradiction.

At least one. Suppose (i) fails. Apply Lemma 4.3.3 to eliminate x_1 , then x_2 from the resulting system, and so on, n times. By part (b) of the lemma, applied n times, the final system has no solution either. But the final system has *no variables*: it is a list of inequalities $0 \leq \beta_k$, which is unsolvable only if some $\beta_k < 0$. By part (a), applied n times, that row $(0, \beta_k)$ is a nonnegative combination of nonnegative combinations of the original rows, hence a nonnegative combination $\sum_i y_i(a_i, b_i)$ with $y \geq 0$ (a product of nonnegative multipliers is nonnegative, and sums of such products are nonnegative). Its coefficient vector is $A^\top y = 0$ and its right-hand side is $b^\top y = \beta_k < 0$. This is (ii). $\square$

Interpretation & discussion

Farkas is a statement about *certificates of infeasibility*: a system of inequalities has no solution exactly when some nonnegative combination of its rows reads $0 \leq$ (negative number). It is the infeasibility twin of weak duality, which produces certificates of optimality. Both say the same thing: for linear inequalities, whatever is true is provable by adding them up with nonnegative weights.

Step 3: the proof of strong duality

Proof of Theorem 4.3.1. By Lemma 4.3.2 let x^* be primal-optimal and set $\gamma = c^\top x^* = p^*$.

Step 1 (what we need). It suffices to find y with $y \geq 0$, $A^\top y \geq c$ and $b^\top y \leq \gamma$. Indeed such a y is dual-feasible, so $b^\top y \geq c^\top x^* = \gamma$ by Theorem 4.2.1; hence $b^\top y = \gamma$, and Theorem 4.2.3 makes y dual-optimal with $d^* = p^*$.

Step 2 (write the requirement as one system). The three conditions form the system $My \leq d$ with

$$M = \begin{bmatrix} -A^\top \\ -I_m \\ b^\top \end{bmatrix} \in \mathbb{R}^{(n+m+1) \times m}, \quad d = \begin{bmatrix} -c \\ 0 \\ \gamma \end{bmatrix}.$$

Step 3 (suppose it is unsolvable and apply Farkas). If the system has no solution, Theorem 4.3.4 gives a vector $(u, v, t) \geq 0$ with $u \in \mathbb{R}^n$, $v \in \mathbb{R}^m$, $t \in \mathbb{R}$ such that $M^\top(u, v, t) = 0$ and $d^\top(u, v, t) < 0$, that is,

$$-Au - v + tb = 0 \quad \text{and} \quad -c^\top u + t\gamma < 0.$$

Rearranged: $Au = tb - v \leq tb$ (since $v \geq 0$) and $c^\top u > t\gamma$.

Step 4 (case $t > 0$: a better plan). Put $\tilde{x} = u/t \geq 0$. Then $A\tilde{x} = b - v/t \leq b$, so $\tilde{x}$ is primal-feasible, and $c^\top \tilde{x} = c^\top u/t > \gamma = p^*$. This contradicts the optimality of x^* .

Step 5 (case $t = 0$: an improving ray). Then $Au = -v \leq 0$, $u \geq 0$ and $c^\top u > 0$. For every $\theta \geq 0$, $x^* + \theta u \geq 0$ and $A(x^* + \theta u) = Ax^* + \theta Au \leq b$, so the whole ray is feasible, and $c^\top u > 0$ makes the objective unbounded along it (Proposition 3.2.4). This contradicts boundedness.

Both cases are impossible, so the system of Step 2 is solvable, and Step 1 finishes the proof. $\square$

Key insight

Farkas turns “no exact certificate exists” into an explicit vector (u, v, t) . Reading that vector back in primal terms gives either a feasible plan better than the optimum ($t > 0$) or a feasible direction of unbounded improvement ($t = 0$). Both contradict the hypotheses, so an exact certificate must exist. The proof is non-constructive; Proposition 4.3.5 below shows where simplex actually finds y^* .

Interpretation & discussion

Strong duality says the bound-finding game is *complete*: the cleverest possible combination of constraints proves the true optimum, never something weaker. It also says the dual is not an auxiliary trick but an equally legitimate description of the same problem; Section 4.4 shows the dual of the dual is the primal. Two readings of the hypotheses matter. First, “bounded objective” is required, not “bounded feasible region”: $\max x_1$ s.t. $x_1 \leq 3, x \geq 0$ in two variables has an unbounded region and a finite optimum, and strong duality applies. Second, if the primal is infeasible or unbounded, the theorem is silent; Section 4.6 covers those cases.

Where simplex finds the dual solution

Strong duality guarantees an exact certificate exists. The simplex method of Chapter 3 hands it to you.

Proposition 4.3.5 (The final objective row is a dual optimal solution). *Run simplex on the equational form $Ax + s = b$, $(x, s) \geq 0$ of (P) , and suppose it stops at a tableau (Definition 3.3.2) in which every reduced cost is ≤ 0 :*

$$v = v_0 + \sum_{j \in N} r_j z_j, \quad z = (x, s), \quad r_j \leq 0.$$

Extend r to all variables by $r_j = 0$ for basic j , and let $r_x \in \mathbb{R}^n$, $r_s \in \mathbb{R}^m$ be the reduced costs of the original and slack variables. Then $y^ := -r_s$ is dual-feasible with $b^\top y^* = v_0$; hence y^* is dual-optimal and the current BFS is primal-optimal.*

Proof. Step 1 (the objective row is an identity). The tableau was obtained from $Ax + s = b$ and $v = c^\top x$ by substitutions that are valid for every (x, s) satisfying the equations, not only for feasible ones. So $c^\top x = v_0 + r_x^\top x + r_s^\top s$ whenever $s = b - Ax$.

Step 2 (substitute the slacks). Replacing s by $b - Ax$, which holds for every $x \in \mathbb{R}^n$ with the corresponding s ,

$$c^\top x = v_0 + r_x^\top x + r_s^\top (b - Ax) = (v_0 + r_s^\top b) + (r_x - A^\top r_s)^\top x \quad \text{for all } x \in \mathbb{R}^n.$$

Step 3 (compare coefficients). Two affine functions of x that agree on all of $\mathbb{R}^n$ have the same constant term and the same linear part: $v_0 + r_s^\top b = 0$ and $r_x - A^\top r_s = c$. With $y^* = -r_s$ these read $b^\top y^* = v_0$ and $A^\top y^* = c - r_x \geq c$, since $r_x \leq 0$. Also $y^* = -r_s \geq 0$. So y^* is dual-feasible with value v_0 , the value of the current BFS, and Theorem 4.2.3 makes both optimal. $\square$

Interpretation & discussion

This is the “seed of duality” noted at the end of Chapter 3: the negatives of the final reduced costs of the slack variables are the optimal dual weights. The juice example there ended with $v = 38 - x_4 - x_5 - x_6$, i.e. $y^* = (1, 1, 1)$. The proposition also explains *why* the optimality test (Theorem 3.3.8) is a proof: a tableau with nonpositive reduced costs *is* a dual-feasible certificate written in a different notation.

Technique: Proof patterns in duality theory

- *Insert the middle term.* To compare $c^\top x$ with $b^\top y$, write $y^\top Ax$ between them and use one sign condition on each side.
- *Equal values $\Rightarrow$ optimal.* Never prove optimality of x directly; exhibit a y and compare values (Theorem 4.2.3).
- *Alternatives.* To prove something *exists*, assume it does not, apply Farkas to get a certificate, and read the certificate as a contradiction (the proof of Theorem 4.3.1).
- *Identities from tableaux.* An objective row is an identity on the whole affine space $\{Az = b\}$; comparing coefficients extracts dual variables (Proposition 4.3.5).

Classic problems

- *The primal has optimal value 19/4. What is the dual optimal value, and why?* 19/4, by strong duality (Theorem 4.3.1); the primal is feasible with a finite maximum.
- *A final tableau reads $v = 38 - x_4 - x_5 - x_6$ where x_4, x_5, x_6 are the slacks. Dual optimal solution? $y^* = (1, 1, 1)$, value 38 (Proposition 4.3.5).*
- *Does strong duality need the feasible region to be bounded?* No; it needs the objective to be bounded above on the region.
- *Show $\{x : x_1 + x_2 \leq 1, -x_1 - x_2 \leq -3\}$ is empty with a Farkas certificate.* $y = (1, 1)$: $A^\top y = 0$, $b^\top y = 1 - 3 = -2 < 0$.

Gotcha

- Strong duality is an *if*: feasible and bounded primal $\Rightarrow$ equal optima. It does not say every LP has a dual with the same value; infeasible or unbounded problems have no value to match.
- “min of the dual equals max of the primal” is about *values*, not solutions: x^* and y^* live in different spaces ($\mathbb{R}^n$ and $\mathbb{R}^m$) and are not transforms of each other.
- Fourier–Motzkin is a proof device, not an algorithm to use by hand: the number of rows can square at every elimination.

Summary: Strong duality

- Feasible and bounded primal $\Rightarrow$ both problems have optimal solutions and $p^* = d^*$ (Theorem 4.3.1).
- Proof route: eliminate variables with nonnegative combinations (Lemma 4.3.3) $\Rightarrow$ an unsolvable system has a certificate $0 \leq \beta < 0$ (Farkas, Theorem 4.3.4) $\Rightarrow$ if no exact dual certificate existed, Farkas would produce a better plan or an improving ray.
- Simplex delivers y^* for free: the negatives of the final reduced costs of the slacks (Proposition 4.3.5).

4.4 Dualization Rules for Any Linear Program

Motivation

Definition 4.1.5 covers only max, $\leq$ rows and ≥ 0 variables. Real models have $\geq$ rows, equalities, free variables and nonpositive variables (Chapter 1). One could convert everything to canonical form first (Propositions 1.3.4 and 1.3.5) and then dualize, but that is slow and error-prone. Better: understand why each sign rule holds and write the dual directly. We first work with a maximisation primal; its dual is a minimisation. Rows become variables, columns become constraints, and the direction of each inequality and the sign of each variable determine the corresponding sign rule.

Every rule below is the answer to one question: “what sign must the multiplier have so that the weighted row still gives an upper bound?”

Rule 1: from row direction to variable sign

We want the weighted row $y_i a_i^\top x$ to be bounded above by $y_i b_i$, so that adding the rows gives an upper bound.

Primal row i	Dual variable	Reason
$a_i^\top x \leq b_i$	$y_i \geq 0$	preserves $\leq$
$a_i^\top x \geq b_i$	$y_i \leq 0$	reverses $\geq$ into $\leq$
$a_i^\top x = b_i$	$y_i \in \mathbb{R}$	an equality survives either sign

For instance, $x_1 + x_2 \geq 3$ multiplied by $y_i = -2$ gives $-2x_1 - 2x_2 \leq -6$, an inequality of the right direction.

Rule 2: from variable sign to row direction

Write $d_j = \sum_i a_{ij} y_i = (A^\top y)_j$ for the weighted coefficient of x_j . We need $c_j x_j \leq d_j x_j$ so that the objective is dominated term by term.

Primal variable j	Dual row j	Reason
$x_j \geq 0$	$d_j \geq c_j$	multiplying by a nonnegative number preserves order
$x_j \leq 0$	$d_j \leq c_j$	multiplying by a nonpositive number reverses order
$x_j \in \mathbb{R}$	$d_j = c_j$	must work for both signs

A free primal variable gives a dual *equality*.

Definition 4.4.1 (General primal–dual pair). Let the primal be $\max c^\top x$ subject to row constraints $a_i^\top x \leq b_i$ ($i \in I_\leq$), $a_i^\top x \geq b_i$ ($i \in I_\geq$), $a_i^\top x = b_i$ ($i \in I_=$), and variable signs $x_j \geq 0$ ($j \in J_\geq$), $x_j \leq 0$ ($j \in J_\leq$), $x_j \in \mathbb{R}$ ($j \in J_{\text{free}}$). Its dual is $\min b^\top y$ subject to $y_i \geq 0$ ($i \in I_\leq$), $y_i \leq 0$ ($i \in I_\geq$), $y_i \in \mathbb{R}$ ($i \in I_=$) and $(A^\top y)_j \geq c_j$ ($j \in J_\geq$), $(A^\top y)_j \leq c_j$ ($j \in J_\leq$), $(A^\top y)_j = c_j$ ($j \in J_{\text{free}}$).

Theorem 4.4.2 (Weak duality for the general pair). For every x feasible for the general primal and every y feasible for its dual, $c^\top x \leq b^\top y$. Consequently feasible x, y with equal values are both optimal.

Proof. **Step 1 (rows).** For each i , $y_i(a_i^\top x - b_i) \leq 0$: if $i \in I_\leq$, the bracket is ≤ 0 and $y_i \geq 0$; if $i \in I_\geq$, the bracket is ≥ 0 and $y_i \leq 0$; if $i \in I_=$, the bracket is 0. Summing over i : $y^\top Ax - y^\top b \leq 0$, i.e. $y^\top Ax \leq b^\top y$.

Step 2 (columns). For each j , $x_j((A^\top y)_j - c_j) \geq 0$: if $j \in J_\geq$, both factors are ≥ 0 ; if $j \in J_\leq$, both are ≤ 0 ; if j is free, the second factor is 0. Summing over j : $x^\top A^\top y - c^\top x \geq 0$, i.e. $c^\top x \leq y^\top Ax$.

Step 3. Chain: $c^\top x \leq y^\top Ax \leq b^\top y$. The optimality statement follows exactly as in Theorem 4.2.3. $\square$

Key insight

The rules are precisely the sign patterns that make each product $y_i(a_i^\top x - b_i)$ nonpositive and each product $x_j((A^\top y)_j - c_j)$ nonnegative. If you forget a rule, rederive it from that requirement.

Theorem 4.4.3 (The general dual is the canonical dual in disguise). Convert the general primal to canonical form by (i) replacing $a_i^\top x \geq b_i$ with $-a_i^\top x \leq -b_i$, (ii) replacing $a_i^\top x = b_i$ with the pair $a_i^\top x \leq b_i$, $-a_i^\top x \leq -b_i$, (iii) substituting $x_j = -x'_j$ with $x'_j \geq 0$ for $j \in J_\leq$, and (iv) substituting $x_j = u_j - w_j$ with $u_j, w_j \geq 0$ for free j . Then the canonical dual of the converted problem is equivalent to the general dual of Definition 4.4.1: feasible solutions correspond with equal objective values. In particular, strong duality (Theorem 4.3.1) holds for the general pair.

Proof. We track what each conversion does to the dual, one at a time; the conversions act on different rows and columns, so they can be composed.

(i) A $\geq$ row. The converted row $-a_i^\top x \leq -b_i$ receives a canonical dual variable $y'_i \geq 0$ that multiplies $-a_i$ and contributes $-b_i y'_i$ to the dual objective. Setting $y_i = -y'_i \leq 0$, the contribution to $A^\top y$ is $y_i a_i$ and to the objective is $b_i y_i$: exactly the general dual with $y_i \leq 0$.

(ii) An equality row. The two converted rows receive $y_i^+, y_i^- \geq 0$, contributing $(y_i^+ - y_i^-)a_i$ to $A^\top y$ and $b_i(y_i^+ - y_i^-)$ to the objective. Every real y_i is of the form $y_i^+ - y_i^-$ (Proposition 1.3.5), and conversely each such difference is a real number: the general dual with $y_i \in \mathbb{R}$.

(iii) A nonpositive variable. Column j becomes $-a^{(j)}$ (where $a^{(j)}$ is column j of A) and the cost becomes $-c_j$; the canonical dual row is $-(a^{(j)})^\top y \geq -c_j$, i.e. $(A^\top y)_j \leq c_j$.

(iv) A free variable. Two columns $a^{(j)}$ and $-a^{(j)}$ with costs c_j and $-c_j$ give the two dual rows $(A^\top y)_j \geq c_j$ and $-(A^\top y)_j \geq -c_j$, together $(A^\top y)_j = c_j$.

The converted primal has the same feasible values as the general primal (Propositions 1.3.4 and 1.3.5), so it is feasible and bounded exactly when the general primal is. Theorem 4.3.1 applies to the converted pair, and the correspondence above transfers optimal solutions and values to the general pair. $\square$

Example 4.4.4 (Mixed signs). Write the dual of

$$\begin{aligned} \max \quad & 3x_1 + x_2 + 4x_3 \\ \text{s.t.} \quad & x_1 + x_2 + x_3 \leq 4, \quad 2x_1 - x_2 + x_3 \geq 1, \quad x_1 + x_3 = 2, \\ & x_1 \geq 0, \quad x_2 \in \mathbb{R}, \quad x_3 \leq 0. \end{aligned}$$

Rows (Rule 1): $y_1 \geq 0, y_2 \leq 0, y_3 \in \mathbb{R}$. Objective: $\min 4y_1 + y_2 + 2y_3$. Columns (Rule 2), keeping every coefficient's sign:

$$\begin{aligned} \text{column of } x_1 = (1, 2, 1) : \quad & y_1 + 2y_2 + y_3 \geq 3 & (x_1 \geq 0), \\ \text{column of } x_2 = (1, -1, 0) : \quad & y_1 - y_2 = 1 & (x_2 \text{ free}), \\ \text{column of } x_3 = (1, 1, 1) : \quad & y_1 + y_2 + y_3 \leq 4 & (x_3 \leq 0). \end{aligned}$$

The dual of the dual

Proposition 4.4.5 (Dualizing twice returns the primal). *The dual of (D), written as a maximisation problem and dualized with the rules above, is equivalent to (P).*

Proof. (D) is $\min b^\top y$ s.t. $A^\top y \geq c, y \geq 0$. By Proposition 1.3.6 it has the same optimal solutions as $\max (-b)^\top y$ s.t. $A^\top y \geq c, y \geq 0$, and by Proposition 1.3.4 the constraints can be written $(-A^\top)y \leq -c$. This is canonical with data $(-A^\top, -c, -b)$, so its dual (Definition 4.1.5) is $\min (-c)^\top x$ s.t. $(-A^\top)^\top x \geq -b, x \geq 0$, i.e. $\min -c^\top x$ s.t. $-Ax \geq -b, x \geq 0$. Reversing the sign of the constraint and of the objective once more gives $\max c^\top x$ s.t. $Ax \leq b, x \geq 0$, which is (P). $\square$

Interpretation & discussion

The two problems are on an equal footing: whichever one you call “primal,” the other is its dual, and the general rules are symmetric under this exchange (a $\geq$ row of a minimisation dualizes to a ≥ 0 variable, and so on). This symmetry is why the primal–dual table of Section 4.6 can be read in both directions.

Technique: The dualization checklist

1. Make sure the primal is a max (if it is a min, either dualize with the mirrored rules or negate the objective first).
2. One dual variable per row; its sign from Rule 1 ($\leq \rightarrow \geq 0, \geq \rightarrow \leq 0, = \rightarrow$ free).
3. Dual objective: $\min b^\top y$ with the primal right-hand sides.
4. One dual constraint per primal variable; read the primal column, keep every sign; its direction from Rule 2 ($\geq 0 \rightarrow \geq c_j, \leq 0 \rightarrow \leq c_j, \text{ free} \rightarrow = c_j$).
5. Always check dimensions, inequality directions and variable signs; then

verify weak duality on one feasible pair as a sanity check.

For a *minimisation* primal the rules mirror: a $\geq$ row gives $y_i \geq 0$, a $\leq$ row gives $y_i \leq 0$; a variable ≥ 0 gives a dual row $\leq c_j$, a variable ≤ 0 gives $\geq c_j$; the dual is a max. Check: applying this to (D) returns (P).

Classic problems

- Dual of $\max c^\top x$ s.t. $Ax = b, x \geq 0$? $\min b^\top y$ s.t. $A^\top y \geq c, y$ free.
- Dual of $\max c^\top x$ s.t. $Ax \leq b, x$ free? $\min b^\top y$ s.t. $A^\top y = c, y \geq 0$.
- Dual of $\min b^\top y$ s.t. $A^\top y \geq c, y \geq 0$? $\max c^\top x$ s.t. $Ax \leq b, x \geq 0$ (Proposition 4.4.5).
- A primal with 5 equality rows and 3 free variables: how many free dual variables and how many dual equalities? 5 free dual variables; 3 dual equalities.

Gotcha

- A $\geq$ row in a *maximisation* gives a *nonpositive* dual variable, not a nonnegative one. Students who “fix” the sign to ≥ 0 produce a wrong dual; verify with Theorem 4.4.2.
- Do not drop the sign of a coefficient when reading a column: in Example 4.4.4 the column of x_2 is $(1, -1, 0)$, giving $y_1 - y_2$.
- A free variable produces an *equality* in the dual, and an equality row produces a *free* variable. The two are easy to swap.
- Converting to canonical form first and then dualizing is correct (Theorem 4.4.3) but produces a dual with split variables and doubled rows; simplify it back before using it.

Summary: Dualization rules

- Rule 1 (rows $\rightarrow$ variables): $\leq \rightarrow y_i \geq 0, \geq \rightarrow y_i \leq 0, = \rightarrow y_i$ free. Rule 2 (variables $\rightarrow$ rows): $x_j \geq 0 \rightarrow (A^\top y)_j \geq c_j, x_j \leq 0 \rightarrow (A^\top y)_j \leq c_j, \text{free} \rightarrow =$.
- Each rule is forced by one requirement: $y_i(a_i^\top x - b_i) \leq 0$ and $x_j((A^\top y)_j - c_j) \geq 0$, which together prove general weak duality (Theorem 4.4.2).
- The general dual equals the canonical dual of the canonical reformulation, so strong duality holds in general (Theorem 4.4.3); the dual of the dual is the primal (Proposition 4.4.5).

4.5 Complementary Slackness

Motivation

Strong duality says $c^\top x^* = b^\top y^*$ at an optimal pair. Weak duality proved this by a chain of two inequalities, $c^\top x \leq y^\top Ax \leq b^\top y$. If the ends agree, both links must be equalities. What does that force on the variables and on the slack in each constraint? The answer, complementary slackness, is both a structural insight and a practical tool: it lets you *compute* a dual solution from a primal one by

At an optimal pair, no resource is both unused and valued, and no activity is both run and overpriced. Every “and” there is a product that must vanish.

solving linear equations.

Slack measures the margin in a constraint

Definition 4.5.1 (Primal and dual slack vectors). For primal-feasible x and dual-feasible y of the canonical pair, define

$$s = b - Ax \geq 0 \quad (\text{primal slack}), \quad r = A^\top y - c \geq 0 \quad (\text{dual slack}).$$

A constraint is *tight* when its slack is 0; a strict inequality has positive slack.

In the workshop, $2x_1 + x_2 \leq 3$ leaves the margin $s_2 = 3 - (2x_1 + x_2)$, the unused capacity of resource 2; the dual constraint $4y_1 + 2y_2 + 3y_3 \geq 2$ leaves the margin $r_1 = 4y_1 + 2y_2 + 3y_3 - 2$, the excess of the weighted cost of activity 1 over its profit.

Lemma 4.5.2 (The gap is a sum of nonnegative products). For primal-feasible x and dual-feasible y ,

$$b^\top y - c^\top x = y^\top (b - Ax) + x^\top (A^\top y - c) = \sum_{i=1}^m y_i s_i + \sum_{j=1}^n x_j r_j,$$

and every term $y_i s_i$ and $x_j r_j$ is nonnegative.

Proof. Insert and remove the middle term: $b^\top y - c^\top x = (b^\top y - y^\top Ax) + (y^\top Ax - c^\top x) = y^\top (b - Ax) + (A^\top y - c)^\top x$, using $y^\top Ax = (A^\top y)^\top x$. Expanding the two inner products gives the sums. Each product is of two nonnegative numbers. $\square$

Theorem 4.5.3 (Complementary slackness). Let x be primal-feasible and y dual-feasible for the canonical pair. Then x and y are both optimal if and only if

$$\begin{aligned} y_i (b_i - (Ax)_i) &= 0 \quad (i = 1, \dots, m), \\ x_j ((A^\top y)_j - c_j) &= 0 \quad (j = 1, \dots, n), \end{aligned}$$

that is, $y_i s_i = 0$ for every i and $x_j r_j = 0$ for every j .

Proof. ($\Rightarrow$) If both are optimal, strong duality (Theorem 4.3.1; the primal is feasible and bounded since it has an optimum) gives $b^\top y - c^\top x = 0$. By Lemma 4.5.2 this is a sum of nonnegative terms equal to zero, so every term is zero: no positive term can be cancelled.

($\Leftarrow$) If every product is zero, Lemma 4.5.2 gives $b^\top y - c^\top x = 0$, i.e. equal objective values. Feasibility and Theorem 4.2.3 then prove both solutions optimal. $\square$

Key insight

A sum of nonnegative numbers is zero only if each is zero. The whole theorem is this one-line fact applied to the decomposition of the duality gap.

Interpretation & discussion

Each variable is paired with a constraint's slack on the *other* side. The conditions read, at an optimal pair:

Primal constraint i : $y_i s_i = 0$	Dual constraint j : $x_j r_j = 0$
Unused margin forces a zero dual weight: $s_i > 0 \Rightarrow y_i = 0$.	A positive activity forces a tight dual constraint: $x_j > 0 \Rightarrow r_j = 0$.
A positive dual weight forces tightness: $y_i > 0 \Rightarrow s_i = 0$.	A positive dual slack forces a zero activity: $r_j > 0 \Rightarrow x_j = 0$.

At least one factor of each product is zero; both may be zero. The economic reading: a resource with spare capacity has no marginal value ($y_i = 0$); an activity that is run must exactly pay for the resources it consumes ($r_j = 0$); an activity whose weighted cost exceeds its profit is not run. Note the direction of the implications: $s_i > 0$ forces $y_i = 0$, but $s_i = 0$ forces nothing about y_i .

Remark 4.5.4 (General pairs). The same statement and proof hold for the general pair of Definition 4.4.1, because the proof of Theorem 4.4.2 showed that each row product $y_i(a_i^\top x - b_i)$ is ≤ 0 and each column product $x_j((A^\top y)_j - c_j)$ is ≥ 0 ; the gap is $\sum_j x_j((A^\top y)_j - c_j) - \sum_i y_i(a_i^\top x - b_i)$, a sum of nonnegative terms. Equality rows and free variables contribute products that are automatically zero.

Using complementary slackness to find a certificate

Example 4.5.5 (From a primal solution to dual weights). Take the workshop point $x = (1/2, 5/4)$ and ask whether it is optimal. Read the slacks:

Resource	Used	Available	Slack
1	12	12	$s_1 = 0$
2	9/4	3	$s_2 = 3/4$
3	4	4	$s_3 = 0$

Step 1 (impose the conditions). If x is optimal and y is a matching dual optimum, Theorem 4.5.3 forces $s_2 > 0 \Rightarrow y_2 = 0$. Both coordinates of x are positive, so $x_1 r_1 = x_2 r_2 = 0$ forces $r_1 = r_2 = 0$: both dual constraints must be tight. With $y_2 = 0$ they become

$$4y_1 + 3y_3 = 2, \quad 8y_1 + 2y_3 = 3.$$

Step 2 (solve). Twice the first equation minus the second: $8y_1 + 6y_3 - 8y_1 - 2y_3 = 4 - 3$, so $4y_3 = 1$ and $y_3 = 1/4$. Substituting, $4y_1 = 2 - 3/4 = 5/4$, so $y_1 = 5/16$.

Step 3 (verify before concluding). $y = (5/16, 0, 1/4) \geq 0$ and $A^\top y = (2, 3)^\top = c$, so y is dual-feasible with $r = (0, 0)$. We also have $s = (0, 3/4, 0)$ and $y_2 = 0$, so $y_1 s_1 = y_2 s_2 = y_3 s_3 = 0$ and $x_1 r_1 = x_2 r_2 = 0$. By Theorem 4.5.3 both solutions are optimal, and $c^\top x = b^\top y = 19/4$.

Interpretation & discussion

The procedure is: read the slacks, impose the conditions, solve, verify feasibility. Step 3 is not optional. Complementary slackness was *assumed* in Step 1 (“if x is optimal”); the equations it produces always have a solution candidate, but the candidate is a certificate only if it is dual-feasible. If the computed y had a negative component or violated $A^\top y \geq c$, the correct conclusion would be “ x is not optimal,” not “the method failed.”

Example 4.5.6 (Detecting a non-optimal point). Try the same procedure on the plan $\hat{x} = (0, 3/2)$. Slacks: $s = (0, 3/2, 1)$, so $y_2 = y_3 = 0$. Only $x_2 > 0$, so only $r_2 = 0$ is forced: $8y_1 = 3$, $y_1 = 3/8$. Candidate $y = (3/8, 0, 0)$. Check dual feasibility: $4 \cdot 3/8 = 3/2 \geq 2$? No. There is no dual-feasible y complementary to $\hat{x}$, so by Theorem 4.5.3 $\hat{x}$ is not optimal, as we already knew ($9/2 < 19/4$).

Technique: Complementary slackness as an exam tool

- *Given a primal candidate x :* compute $s = b - Ax$. Every $s_i > 0$ sets $y_i = 0$; every $x_j > 0$ sets $(A^\top y)_j = c_j$. Solve the resulting linear system for the remaining y_i ; verify $y \geq 0$ and $A^\top y \geq c$. Feasible: x optimal and y its certificate. Infeasible or inconsistent: x not optimal.
- *Given a dual candidate y :* symmetric. Every $r_j > 0$ sets $x_j = 0$; every $y_i > 0$ makes row i tight.
- *Counting equations:* with k positive primal variables and ℓ positive slacks, you get k equations in $m - \ell$ unknowns. For a nondegenerate vertex, $k = m - \ell$ and the system is square.
- *Nothing is forced* by $s_i = 0$ or $x_j = 0$. Do not write $y_i > 0$ because resource i is fully used; it may still have $y_i = 0$ (degenerate case).

Classic problems

- *For max $3x_1 + 2x_2$ s.t. $x_1 + x_2 \leq 4$, $2x_1 + x_2 \leq 6$, $x \geq 0$:* is $x = (2, 2)$ optimal? Both rows tight, both variables positive: $y_1 + 2y_2 = 3$, $y_1 + y_2 = 2$, so $y = (1, 1) \geq 0$. Feasible, so yes; value 10.
- *At an optimal pair, $y_3 = 2 > 0$. What do you know about resource 3?* It is fully used: $s_3 = 0$.
- *At an optimal pair, $x_2 = 0$. Is dual row 2 tight?* Not necessarily: $x_2 r_2 = 0$ holds whatever r_2 is.
- *Is x optimal if some complementary y exists with $A^\top y \geq c$ but $y_1 < 0$?* No: y is not dual-feasible, so it certifies nothing; try again or conclude non-optimality if the system forces $y_1 < 0$.

Gotcha

- Complementary slackness is a condition on a *pair*: “ x satisfies complementary slackness” means nothing by itself.
- The implications run one way: positive slack kills the dual variable; zero slack does *not* make it positive. Degenerate vertices have tight

constraints with zero multipliers.

- Skipping the verification step is the classic error. The equations assume optimality; the feasibility check tests it.

Summary: Complementary slackness

- $b^\top y - c^\top x = \sum_i y_i s_i + \sum_j x_j r_j$ with $s = b - Ax$, $r = A^\top y - c$ (Lemma 4.5.2).
- Feasible x, y are both optimal iff $y_i s_i = 0$ for all i and $x_j r_j = 0$ for all j (Theorem 4.5.3).
- Reading: spare capacity $\Rightarrow$ zero weight; positive activity $\Rightarrow$ tight dual row. Use it to compute a certificate from a candidate, then *verify feasibility*.

4.6 Three Outcomes and the Primal–Dual Table

Motivation

Not every LP has a finite optimum. This section classifies what can happen to one LP and then to a primal–dual pair, using nothing but weak and strong duality. The classification is exhaustive and each cell of the table is either proved possible by an example or proved impossible by a theorem.

Strong duality handles the “nice” case. The table tells you what the dual looks like when the primal is not nice.

Proposition 4.6.1 (Three outcomes). *For a maximisation LP, exactly one of the following occurs:*

1. Finite optimum: *a feasible solution attains a finite best objective value;*
2. Unbounded objective: *feasible solutions exist and their objective values grow without limit;*
3. Infeasible: *no point satisfies all the constraints.*

For a minimisation, “unbounded” means the objective can decrease without limit.

Proof. The feasible set is empty (case 3) or not. If not, the objective is either unbounded above on it (case 2) or bounded above. In the latter case the optimum is *attained* by Lemma 4.3.2 (via Theorem 2.5.2, after conversion to equational form), which is case 1. The three cases are mutually exclusive by definition. $\square$

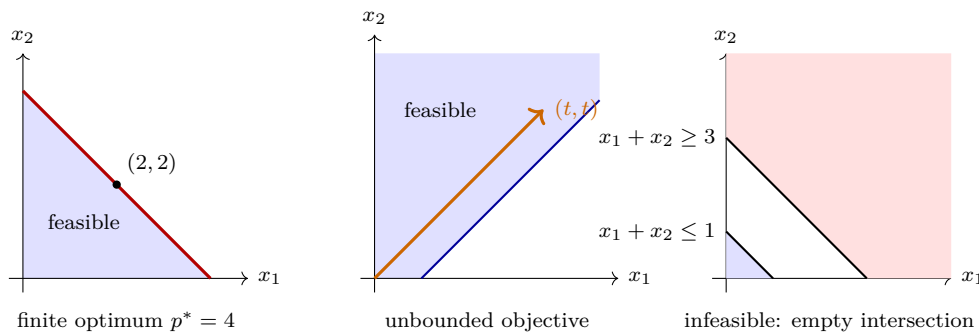
Interpretation & discussion

“Bounded, hence attained” is special to linear programming. For a general optimisation problem, sup need not be a max ($\max -e^{-x}$ over $x \geq 0$ has supremum 0, never attained). For LPs, Theorem 2.5.2 rules this out: the finitely many basic feasible solutions include an optimal one.

Example 4.6.2 (Finite optimum). $\max x_1 + x_2$ s.t. $x_1 + x_2 \leq 4$, $x \geq 0$. Every feasible value is at most 4 (the constraint itself is the certificate: $y = 1$). The feasible point $(2, 2)$ attains it: $p^* = 4$. The optimum is not unique: every point of the segment from $(4, 0)$ to $(0, 4)$ has value 4.

Example 4.6.3 (Unbounded objective). $\max x_1 + x_2$ s.t. $x_1 - x_2 \leq 1, x \geq 0$. For every $t \geq 0$ the point (t, t) is feasible since $t - t = 0 \leq 1$, and $x_1 + x_2 = 2t \rightarrow +\infty$. There is no finite maximum (Proposition 3.2.4 with $x^0 = 0, d = (1, 1)$). An unbounded feasible set alone does not imply an unbounded objective: the same region with objective $\max x_1 - x_2$ has optimum 1.

Example 4.6.4 (Infeasible). $\max x_1 + x_2$ s.t. $x_1 + x_2 \leq 1, x_1 + x_2 \geq 3, x \geq 0$. A feasible point would satisfy $3 \leq x_1 + x_2 \leq 1$, impossible. The feasible set is empty. (Farkas certificate: add the two rows written as $x_1 + x_2 \leq 1$ and $-x_1 - x_2 \leq -3$ with weights $(1, 1)$: $0 \leq -2$.)



What can happen to a primal–dual pair?

Theorem 4.6.5 (Primal–dual outcome table). *For the canonical pair (and, by Theorem 4.4.3, for any general pair), the possible combinations are exactly:*

Primal (maximisation)	Dual (minimisation)
finite optimum	finite optimum, same value
unbounded above	infeasible
infeasible	unbounded below
infeasible	infeasible

In particular: if one problem is unbounded, the other is infeasible; if one has a finite optimum, so does the other; and an infeasible problem's dual is either unbounded or infeasible, never finite.

Proof. There are nine combinations of the three outcomes on each side; we show which survive.

Finite primal $\Rightarrow$ finite dual, same value. Theorem 4.3.1. By symmetry (Proposition 4.4.5 and Theorem 4.4.3), finite dual $\Rightarrow$ finite primal. This settles the first row and rules out (finite, unbounded), (finite, infeasible), (unbounded, finite), (infeasible, finite).

Unbounded primal $\Rightarrow$ infeasible dual. If the dual had a feasible y , weak duality (Theorem 4.2.1) would give $c^\top x \leq b^\top y$ for every feasible x , a finite upper bound, contradicting unboundedness. Symmetrically, unbounded dual $\Rightarrow$ infeasible primal. This rules out (unbounded, unbounded) and gives rows two and three as the only options when one side is unbounded.

The remaining cells occur. (unbounded, infeasible): Example 4.6.3; its dual $\min y_1$ s.t. $y_1 \geq 1, -y_1 \geq 1, y_1 \geq 0$ is infeasible. (infeasible, unbounded): the

dual of that example, read as primal (Proposition 4.4.5). (infeasible, infeasible): Example 4.6.6 below. $\square$

Example 4.6.6 (Both problems infeasible).

Primal: $\max x_1 + x_2$ s.t. $x_1 - x_2 \leq -1$, $-x_1 + x_2 \leq -1$, $x \geq 0$;

Dual: $\min -y_1 - y_2$ s.t. $y_1 - y_2 \geq 1$, $-y_1 + y_2 \geq 1$, $y \geq 0$.

Adding the primal constraints gives $0 \leq -2$; adding the dual constraints gives $0 \geq 2$. Neither side has a feasible point, and there is no finite optimum to compare.

Interpretation & discussion

Infeasibility of one problem does *not* automatically mean the other is unbounded; Example 4.6.6 shows both can fail. What *is* automatic: an unbounded problem forces its partner to be infeasible, because a single feasible point on the other side would be a finite bound. The table is the complete list of what duality allows.

A complete worked certificate

Example 4.6.7 (Prove an optimum using duality). Consider $\max 3x_1 + 2x_2$ s.t. $x_1 + x_2 \leq 4$, $2x_1 + x_2 \leq 6$, $x \geq 0$.

1. *Dual*: $\min 4y_1 + 6y_2$ s.t. $y_1 + 2y_2 \geq 3$, $y_1 + y_2 \geq 2$, $y \geq 0$.
2. *Primal feasibility* of $x = (2, 2)$: $2 + 2 = 4 \leq 4$, $4 + 2 = 6 \leq 6$, $x \geq 0$. Value $c^\top x = 10$, so $p^* \geq 10$.
3. *A matching certificate*: both primal rows are tight and both variables positive, so complementary slackness suggests making both dual rows tight: $y_1 + 2y_2 = 3$, $y_1 + y_2 = 2$, giving $y = (1, 1)$. Check: $y \geq 0$, $1 + 2 = 3 \geq 3$, $1 + 1 = 2 \geq 2$. Value $b^\top y = 4 + 6 = 10$, so $p^* \leq 10$.
4. *Conclusion*: feasible x , feasible y , equal values, so both are optimal and $p^* = 10$ (Theorem 4.2.3, i.e. weak duality). The certificate is visible by adding the primal rows: $(x_1 + x_2) + (2x_1 + x_2) \leq 4 + 6$, i.e. $3x_1 + 2x_2 \leq 10$.

Technique: Classifying an LP and its dual quickly

- A feasible improving ray ($d \geq 0$, $Ad \leq 0$, $c^\top d > 0$ in canonical form) proves unboundedness and, at once, dual infeasibility.
- A nonnegative combination of rows reading $0 \leq$ negative proves infeasibility (Farkas); do not then assume the dual is unbounded, check it.
- One feasible point on each side proves both have finite optima (weak duality bounds both, Theorem 2.5.2 attains them).
- If you can find a feasible dual point, the primal cannot be unbounded, whatever else is true.

Classic problems

- *The primal is unbounded. Dual?* Infeasible.
- *The dual is infeasible. Primal?* Unbounded or infeasible; more information needed.
- *Both problems have feasible points. What follows?* Both have finite optima with equal values.
- $\max x_1 + x_2$ s.t. $x_1 - x_2 \leq 1, x \geq 0$: write the dual and classify. Dual: $\min y_1$ s.t. $y_1 \geq 1, -y_1 \geq 1, y_1 \geq 0$: infeasible, consistent with the unbounded primal.

Gotcha

- “Unbounded feasible region” and “unbounded objective” are different statements; only the second is an outcome of the LP.
- “Infeasible $\Rightarrow$ partner unbounded” is false (Example 4.6.6). “Unbounded $\Rightarrow$ partner infeasible” is true.
- A bounded LP attains its optimum (Proposition 4.6.1); do not write “supremum” for an LP unless it is infeasible or unbounded.

Summary: Outcomes and the primal–dual table

- An LP is infeasible, unbounded, or has an attained finite optimum (Proposition 4.6.1).
- Finite $\leftrightarrow$ finite with equal values; unbounded $\Rightarrow$ partner infeasible; infeasible $\Rightarrow$ partner unbounded or infeasible (Theorem 4.6.5).
- The full verification recipe: write the dual, check primal feasibility, find a dual-feasible point with the same value (complementary slackness tells you which rows to make tight), conclude by weak duality.

Notes on the original slides

Every number in the Lecture 4 slides was recomputed: the three bounds 6, 5, $19/4$ for the workshop, the feasibility of $\hat{x} = (0, 3/2)$ and of $x^* = (1/2, 5/4)$, the dual feasibility of $(1/3, 1/3, 0)$ and $(5/16, 0, 1/4)$, the complementary-slackness derivation of y^* , the mixed-signs dual of slide 23, the dual-of-dual computation, the examples of the three outcomes, the both-infeasible pair, and the $(2, 2), (1, 1)$ certificate of slides 39–41. All are correct. Points worth a reader’s attention:

- **Strong duality (slide 18)** is stated without proof. Section 4.3 proves it from Fourier–Motzkin elimination and the Farkas lemma, following the route of Matouek and Gärtner, Chapter 6, and adds the constructive reading via the final simplex tableau (Proposition 4.3.5).
- **“Feasibility + equal values $\Rightarrow$ both optimal” (slide 17)** is a consequence of weak duality alone (Theorem 4.2.3); strong duality is only needed to know that such a pair *exists*.
- **Complementary slackness (slide 27)** is stated for the canonical pair. The “ $\Rightarrow$ ” direction uses strong duality; the “ $\Leftarrow$ ” direction uses only weak

duality. The remark after Theorem 4.5.3 extends it to general pairs.

- **Slide 32 (“read the slacks, impose, solve, verify”).** The verification step is essential: Example 4.5.6 shows the same procedure applied to a non-optimal point produces an infeasible candidate, which is how non-optimality is detected.
- **Three outcomes (slide 33).** “Exactly one occurs” silently uses that a bounded LP *attains* its optimum, which is Theorem 2.5.2; for non-linear problems this can fail.
- **Slide 37, “infeasible does not automatically mean the other is unbounded.”** Correct; the table’s fourth row is realised by slide 38, reproduced as Example 4.6.6. The missing cells (finite with non-finite, unbounded with unbounded) are impossible by Theorem 4.6.5.
- **Slide 24 (dual of the dual)** uses only the canonical rules plus $\min \leftrightarrow \max$ and sign reversal; the general dualization rules (slides 20–21) are proved in Theorems 4.4.2 and 4.4.3.

Chapter takeaways

Summary: Duality of linear programs

- **Two searches:** the primal searches for a good feasible decision; the dual searches for a good bound on what is possible. Rows become variables, columns become constraints.
- **Weak duality:** feasible decisions never beat valid bounds; a feasible pair with equal values is optimal, and that is a two-line proof.
- **Strong duality:** at a finite LP optimum the best bounds meet; simplex finds the exact certificate in its final objective row.
- **Dualization rules:** $\leq / \geq / =$ rows give $\geq 0 / \leq 0 /$ free variables; $\geq 0 / \leq 0 /$ free variables give $\geq / \leq / =$ rows. Each rule is forced by one sign requirement.
- **Complementary slackness:** spare capacity means zero weight, positive activity means tight dual row; use it to compute a certificate, then verify.
- **Outcomes:** finite pairs with finite, unbounded pairs with infeasible, and infeasible may pair with either; always check dimensions, directions and signs.